Data Cleaning

- Raw data is often noisy, incomplete and inconsistent --> impact the model performance
- Goal: to ensure that the data is accurate, consistent and free of errors, complete data, and quality.
- Clean data = easy to interpret and important in EDA (Exploratory Data Analysis)

Import library and check the data

```
import pandas as pd
import numpy as np

df = pd.read_csv('Titanic-Dataset.csv')
df.info()
df.head()

df.duplicated()
```

Identify Data types

```
cat_col = [col for col in df.columns if df[col].dtype == 'object']
num_col = [col for col in df.columns if df[col].dtype != 'object']

print('Categorical columns:', cat_col)
print('Numerical columns:', num_col)
```

Count unique values in the categorial Columns

```
df[cat_col].nunique()
```

---> The output will look like this

	0
Name	891
Sex	2
Ticket	681
Cabin	147
Embarked	3

dtype: int64

Detect missing values and return the boolean expression

```
df.isnull()
```

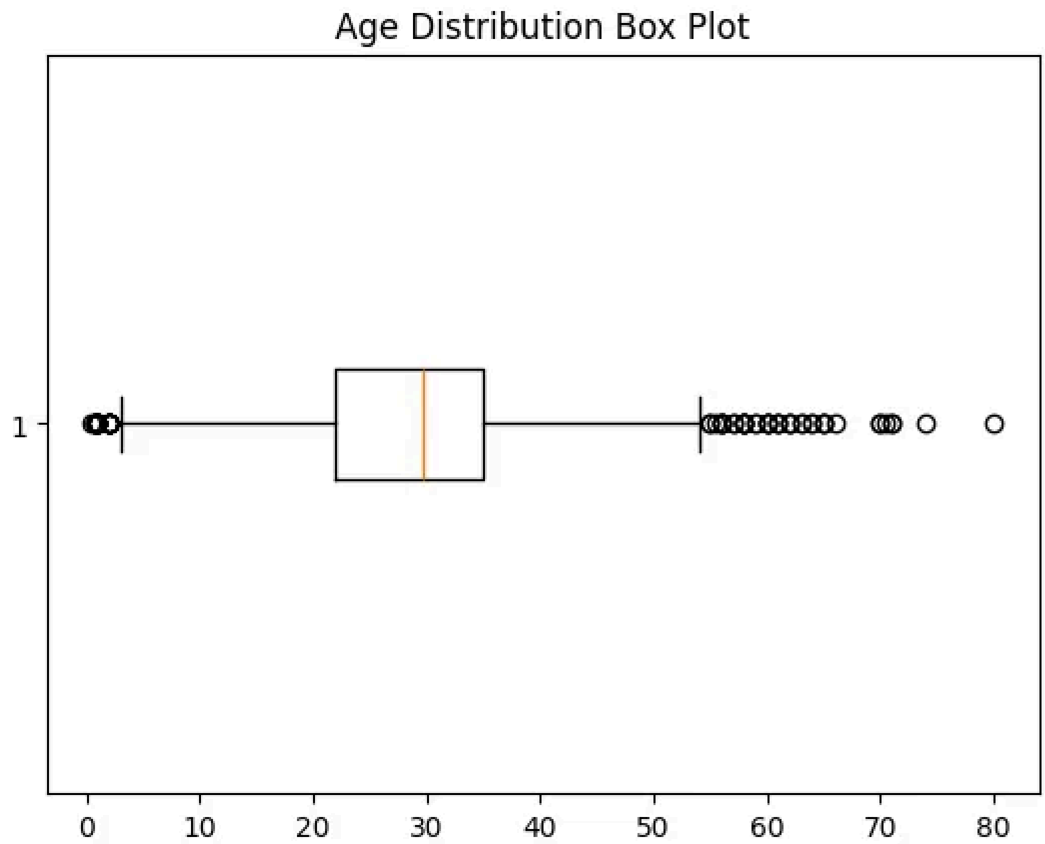
Drop Irrelevant or Data-heavy Missing Columns

- df.drop(columns=[]): Drops specified columns from the DataFrame.
- df.dropna(subset=[]): Removes rows where specified columns have missing values.
- fillna(): Fills missing values with specified value (e.g., mean).
  Detect Outlier with boxplot

```
import matplotlib.pyplot as plt

plt.boxplot(df3['Age'], vert=False)
plt.ylabel('Variable')
plt.xlabel('Age')
plt.title('Box Plot')
plt.show()
```

---> **The output**



Calculate outlier Boundaries and Remove them

```
mean = df1['Age'].mean()
std = df1['Age'].std()
```

```
lower_bound = mean - 2 * std
upper_bound = mean + 2 * std

df2 = df1[(df1['Age'] >= lower_bound) & (df1['Age'] <= upper_bound)]
```

Impute Missing data again if any issues arise

```
]df3 = df2.fillna(df2['Age'].mean())
df3.isnull().sum()
```